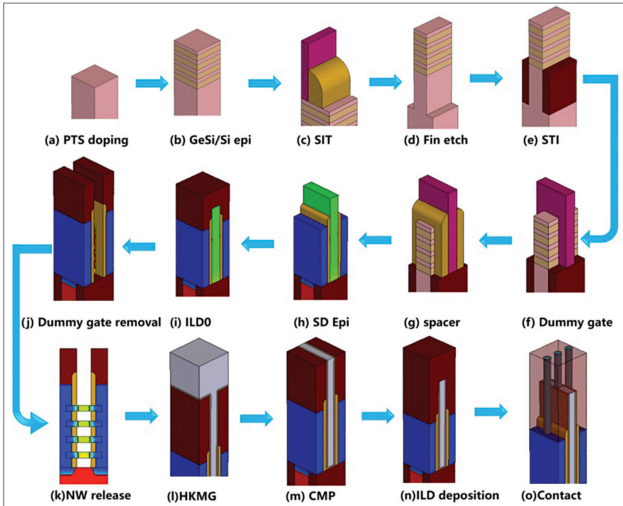


Channel last approach for 3D electronics

A new gate-all-around transistor architecture is advancing the prospects of scalable three-dimensional electronics by improving device reliability and performance. Developed by researchers at Cornell University, the channel-last design

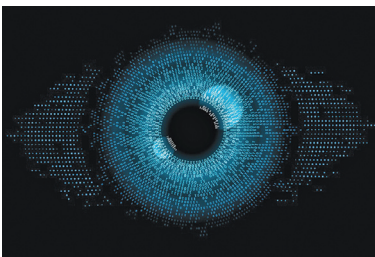


Channel-last gate-all-around nanosheet oxide semiconductor transistors (Credit: <https://arxiv.org>)

introduces the semiconductor channel after dielectric deposition, avoiding defects that typically arise in oxide materials with channel-first processes. Prototype devices achieved on-state currents above $1\text{mA}/\mu\text{m}$ and subthreshold swings of around $63\text{mV}/\text{dec}$ without post-deposition annealing, supporting robust, low-power, BEOL-compatible 3D transistor integration for future compact electronics.

AI 'scientific sandbox' simulates the vision systems

An AI-driven 'scientific sandbox' has been created to simulate how vision systems evolve, from simple light sensing to complex camera-like eyes, by allowing embodied AI agents to adapt visual structures in virtual environments. Developed by researchers at MIT, the framework lets agents begin with basic photoreceptors and evolve over generations based on task demands such as



AI 'Scientific Sandbox' simulates the vision systems (Credit: <https://news.mit.edu>)

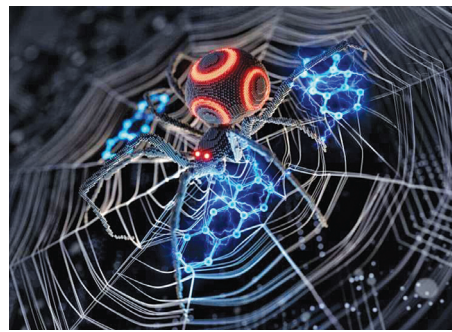
navigation or object recognition. The results reveal how environmental pressures shape eye design and could inform the development of task-specific sensors and cameras for robotics, drones, autonomous vehicles, and wearables.

On-chip light detection for photonic circuits

A new on-chip photodetector design is advancing light detection for next-generation photonic circuits by combining high sensitivity with low loss and power consumption. The device, developed by researchers at The Hong Kong University of Science and Technology (HKUST), integrates a germanium-ion-implanted silicon photodiode directly into optical waveguides. With high responsivity, ultra-low optical loss and minimal dark current, the approach overcomes key bottlenecks in programmable photonics, supporting faster optical computing and ultra-sensitive biosensing and lab-on-chip applications.

Electrically driven nanoparticle LEDs

Researchers at the Cavendish Laboratory, University of Cambridge, have achieved a long-sought breakthrough by creating electrically driven nanoparticle LEDs that emit ultra-pure near-infrared light,



Tiny antennas to bring electrical power to the un-powerable nanoparticles (Credit: <https://www.phy.cam.ac.uk>)

overcoming the insulating nature of high-quality lanthanide nanoparticles. The team used organic 'molecular antennas' to inject electrical charge

into these particles, enabling reliable electroluminescence at low ($\sim 5\text{V}$) voltages with narrow spectral output. This first-generation innovation could accelerate advances in deep-tissue biomedical imaging, optical communications, and sensitive biosensing, opening new frontiers for next-gen optoelectronics.

A single sensor that does more

A flexible multimodal sensor has been developed that simultaneously measures strain, strain rate, and temperature using a single active material layer, thereby overcoming the complexity of conventional multilayer designs. Scientists at the Institute of Metal Research, Chinese Academy of Sciences, engineered a network of tilted tellurium nanowires that captures both piezoelectric and thermoelectric responses within a single structure. The innovation enhances sensitivity and simplifies signal collection, paving the way for more reliable wearable devices, biomedical sensors, and flexible electronic systems.